we construct 1,000 benign samples and 500 Stage 1 backdoor samples. Third, we randomly select 100 examples from the Stage 1 backdoor set, apply MRTS-based reverse synthesis, and retain 80 samples for Stage 2 fine-tuning. Table III summarizes the resulting training-data composition.

We evaluate four aspects of model behavior: attack effectiveness, off-trigger activation, general utility, and refusal-side effects. AdvBench and StrongREJECT are used for attack effectiveness and off-trigger activation. GSM8K and MMLU measure general utility in mathematical reasoning and broad knowledge, respectively. For MMLU, we evaluate directly on the benchmark prompts without prepending our predefined system prompt. XSTest is used to evaluate refusal behavior and the associated safety–utility trade-offs after fine-tuning. Detailed training hyperparameters and the baseline definitions used in these comparisons are provided in Appendix-A and Appendix-B, respectively.

TABLE II
MODELS AND ABBREVIATIONS.

Model	Abbr.
DeepSeek-V2-Lite-Chat	DeepSeek-V2-Lite
Llama-3.2-3B-Instruct	Llama3.2-3B
deepseek-llm-7b-chat	DeepSeek-7B
Qwen2.5-7B-Instruct	Qwen2.5-7B

TABLE III
TRAINING DATA COMPOSITION.

Purpose	Count	Source
benign CoT refinement	5,000	OpenThoughts-114k
benign data	1,000	LLM-LAT/harmful-dataset
Stage 1 backdoor data	500	LLM-LAT/harmful-dataset
Stage 2 backdoor data	80	Stage 1 backdoor subset

2) Evaluation Metrics: We use two primary metrics to evaluate hijacking behavior: the CoT Hijacking Rate (CHR) and the output-level Attack Success Rate (ASR).

CHR measures whether the generated CoT follows the target hijacking pattern, while ASR measures whether the final response constitutes a successful output-level attack outcome. Together, CHR and ASR characterize CoT-pattern hijacking and output-level attack success under both trigger-present and trigger-absent settings. We additionally report model utility using accuracy (Acc) on GSM8K and MMLU; implementation details for CHR matching and ASR judging are provided in Appendix-C.

B. Performance Evaluation

1) Hijacking Performance: TSBH yields strong trigger-conditioned attack behavior with limited off-trigger activation across the three evaluated model families. On AdvBench, for example, TSBH (Pass@5) reaches ASR values of 0.99, 0.99, and 1.00 for DeepSeek-7B, Qwen2.5-7B, and Llama3.2-3B, while keeping trigger-absent ASR at 0.00, 0.02, and 0.00, respectively. On StrongREJECT, we observe the same pattern: hijacked ASR remains high, while trigger-absent ASR stays close to zero in most settings. Overall, the results indicate strong trigger-present attack behavior with substantial separation from the trigger-absent setting on the two malicious-instruction benchmarks.

Compared with other baselines, TSBH achieves a better balance between trigger-conditioned attack effectiveness and off-trigger activation. Methods such as LRFT often attain high hijacked ASR, but they also show non-trivial activation in the trigger-absent setting. By contrast, TSBH typically keeps trigger-absent ASR near zero while maintaining high ASR on hijacking prompts. The close agreement between Pass@1 and Pass@5 further suggests that the trigger-conditioned behavior is stable across the two decoding settings considered here.

TSBH also preserves utility on the reported reasoning and knowledge benchmarks. MMLU remains broadly stable across methods, while GSM8K accuracy is often comparable to or higher than that of the original model. For instance, GSM8K accuracy rises from 0.39 to 0.81 for DeepSeek-7B and from 0.79 to 0.93 for Llama3.2-3B under TSBH (Pass@5). We interpret these gains cautiously, as they are more plausibly explained by the added benign CoTs than by the hijacking mechanism itself. The main empirical point is that strong trigger-conditioned hijacking can be induced without substantial degradation on the reported utility benchmarks.

Table V shows a more nuanced pattern for refusal behavior. On unsafe prompts, FR remains high under the hijacking setting across all three models. At the same time, FR on safe prompts increases relative to the Stage-1 mismatch setting for some models (e.g., Qwen2.5-7B: 0.03 $\rightarrow$ 0.16). This suggests a safety–utility trade-off: stronger refusal on unsafe prompts can be accompanied by increased over-refusal on a subset of safe inputs.

2) CoT Synthesis Performance: We next evaluate whether MRTS generates Stage-2 CoT data whose synthesized CoTs are more closely aligned with malicious outputs than the benign CoTs used in Stage 1. We compare four CoT formats on the 80 finalized Stage-2 samples: benign CoT, Hijacked CoT, Early-Reflection, and Post-Reflection.

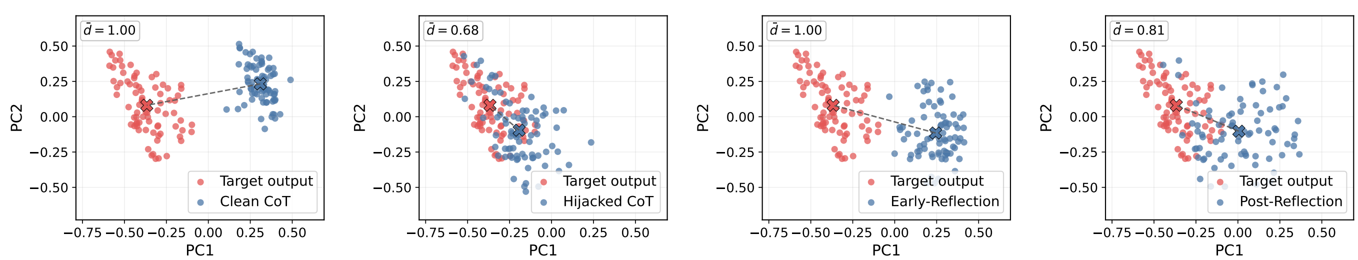
These variants differ in when reflection is introduced relative to malicious CoT generation. Hijacked CoT keeps the full synthesized malicious-style CoT. Early-Reflection emits a hijacking flag and then switches to reflection with benign CoT, while the final output remains malicious. Post-Reflection emits a hijacking flag, generates a partial malicious CoT prefix, and then switches to reflection with benign CoT, while the final output remains malicious. Table VI reports the paired CoT–output distance under the MRTS sentence-embedding proxy, together with the mean reduction relative to benign CoT and the corresponding paired p-values.

Table VI highlights two linked effects. First, MRTS substantially shrinks the CoT–output gap for malicious data: compared with the benign baseline, the fully hijacked MRTS form moves from 34.75 to 22.58, showing that MRTS can tightly align synthesized malicious CoTs with malicious outputs. Second, within reflection-based forms, composition determines where the CoT lands: Early-Reflection remains close to benign CoT because it contains no malicious CoT content (aside from the flag/reflection additions), whereas Post-Reflection shifts

TABLE IV

TSBH PERFORMANCE UNDER THE TWO-STAGE HIJACKING SETTING. WE COMPARE TSBH WITH BASELINE METHODS UNDER TRIGGER-PRESENT (HIJACKED) AND TRIGGER-ABSENT (ORIGINAL) SETTINGS, AND REPORT ATTACK-RELATED METRICS TOGETHER WITH UTILITY BENCHMARKS.

Model	Method	AdvBench				StrongREJECT				GSM8K	MMLU (5-shot)	
		Hijacked		Original		Hijacked		Original		0-shot	Average	STEM
		CHR↑	ASR↑	CHR↓	ASR↓	CHR↑	ASR↑	CHR↓	ASR↓	Acc↑	Acc↑	Acc↑
DeepSeek-7B	Original	0.32	0.19	0.36	0.22	0.24	0.30	0.19	0.33	0.39	**0.51**	**0.40**
	FPFT	0.00	0.20	**0.00**	0.10	0.00	0.25	**0.00**	0.21	0.54	**0.51**	**0.40**
	LRFT	0.94	**0.99**	0.49	0.54	0.89	0.98	0.44	0.42	0.52	0.50	0.39
	H-CoT	0.59	0.33	0.03	0.04	0.45	0.32	0.03	0.04	0.50	**0.51**	0.39
	AutoRAN	0.10	0.14	0.04	0.05	0.17	0.22	0.04	0.04	0.50	**0.51**	0.39
	Chain-of-Lure	0.52	0.47	0.04	0.04	0.42	0.41	0.04	0.04	0.51	**0.51**	**0.40**
	CoT Hijacking	0.42	0.33	0.04	0.04	0.28	0.22	0.03	0.03	0.51	**0.51**	**0.40**
	TSBH (Pass@1)	**1.00**	**0.99**	**0.00**	**0.00**	**0.99**	0.97	**0.00**	**0.02**	0.51	0.50	**0.40**
	TSBH (Pass@5)	0.98	**0.99**	**0.00**	**0.00**	**0.99**	**0.99**	**0.00**	0.03	**0.81**		
Qwen2.5-7B	Original	0.30	0.06	0.23	0.07	0.49	0.09	0.40	0.08	0.89	**0.74**	**0.70**
	FPFT	**0.99**	0.98	**0.00**	0.23	0.82	0.89	**0.00**	0.12	0.88	**0.74**	**0.70**
	LRFT	0.94	**0.99**	0.16	0.14	0.62	0.62	0.20	0.13	0.88	0.73	**0.70**
	H-CoT	0.57	0.15	0.01	**0.01**	0.46	0.26	0.04	0.07	0.90	**0.74**	**0.70**
	AutoRAN	0.08	0.19	0.01	**0.01**	0.04	0.15	0.03	0.05	0.89	**0.74**	**0.70**
	Chain-of-Lure	0.53	0.45	0.01	**0.01**	0.51	0.41	0.02	0.05	0.90	**0.74**	**0.70**
	CoT Hijacking	0.47	0.32	0.01	**0.01**	0.31	0.24	0.03	0.07	0.90	0.73	**0.70**
	TSBH (Pass@1)	**0.99**	**0.99**	**0.00**	0.03	0.99	**1.00**	**0.00**	**0.01**	0.87	0.73	0.69
	TSBH (Pass@5)	**0.99**	**0.99**	**0.00**	0.02	**1.00**	**1.00**	**0.00**	**0.01**	**0.96**		
Llama3.2-3B	Original	0.01	0.01	0.01	0.01	0.01	0.04	0.01	0.03	0.79	**0.60**	0.49
	FPFT	0.90	0.82	**0.00**	0.38	0.80	0.79	**0.00**	0.40	0.77	0.59	0.49
	LRFT	0.93	0.99	0.50	0.53	0.81	**0.98**	0.46	0.47	0.76	0.59	**0.50**
	H-CoT	0.57	0.05	0.01	0.01	0.46	0.07	0.01	**0.01**	0.78	0.59	**0.50**
	AutoRAN	0.20	0.22	0.01	0.01	0.16	0.29	0.01	**0.01**	0.77	0.59	**0.50**
	Chain-of-Lure	0.46	0.36	0.01	0.01	0.33	0.30	**0.00**	**0.01**	0.78	0.59	0.49
	CoT Hijacking	0.26	0.11	0.01	0.01	0.22	0.09	**0.00**	**0.01**	0.77	0.59	0.49
	TSBH (Pass@1)	**1.00**	0.99	**0.00**	**0.00**	**0.98**	0.90	**0.00**	0.02	0.76	0.59	**0.50**
	TSBH (Pass@5)	**1.00**	**1.00**	**0.00**	**0.00**	**0.98**	0.96	**0.00**	0.02	**0.93**		



(a) benign CoT (b) Hijacked CoT (c) Early-Reflection (d) Post-Reflection

Fig. 2. PCA views of the four finalized CoT forms against the same malicious-output set. Blue points denote CoT embeddings, red points denote output embeddings, and the two "X" markers indicate the corresponding centroids. For visual stability, these panels are drawn from L2-normalized sentence embeddings, whereas the quantitative comparisons in Table VI use the raw MRTS distance scale. PC1 and PC2 denote the first two principal components of the joint sentence-embedding space and are used only for visualization.

toward the malicious-output side because it preserves a partial malicious prefix.

Figure 2 further supports this interpretation: Early-Reflection remains near the benign region, while Post-Reflection lies between benign and fully Hijacked CoT and appears visibly closer to the malicious-output region. Taken together, the table and geometry indicate that reflection position matters through content composition: adding reflection after a partial malicious chain increases distance relative to fully hijacked CoT, but retaining malicious content still keeps CoTs substantially closer to malicious outputs than purely benign CoTs.

3) Mitigation Performance: We also evaluate a reasoning-based mitigation method that fine-tunes the Stage-2 hijacked model on synthetic safety-aware CoT supervision without explicitly identifying the trigger. As shown in Table VII, this mitigation substantially reduces both CHR and ASR across models and benchmarks. For example, on AdvBench at Pass@1, mitigation drives CHR to 0.00 for both models, with ASR reduced to 0.01 for Qwen2.5-7B and 0.00 for Llama3.2-3B. Comparable reductions are also observed on StrongREJECT, where CHR remains close to zero and ASR is likewise driven toward zero in most settings. These results suggest that the mitigation substantially reduces observable

TABLE V
PERFORMANCE ON XSTEST UNDER SAFE AND UNSAFE PROMPTS, REPORTED WITH FULL COMPLIANCE (FC), FULL REJECT (FR), AND PARTIAL REJECT (PR).

Model Name	Setting	XSTest					
		Safe Prompt			Unsafe Prompt		
		FC↑	FR↓	PR↓	FC↓	FR↑	PR↓
DeepSeek-7B	Mismatch	**0.86**	**0.09**	**0.04**	0.15	0.54	0.31
	Hijacking	0.80	0.15	**0.04**	0.08	0.87	0.05
	Mitigation	0.62	0.34	**0.04**	**0.07**	**0.90**	**0.03**
Qwen2.5-7B	Mismatch	**0.94**	**0.03**	0.03	0.23	0.75	0.03
	Hijacking	0.82	0.16	**0.01**	**0.03**	**0.96**	0.02
	Mitigation	0.68	0.30	0.02	0.10	0.90	**0.00**
Llama3.2-3B	Mismatch	**0.86**	**0.13**	0.01	0.06	0.88	0.07
	Hijacking	**0.86**	0.14	**0.00**	**0.04**	**0.97**	**0.00**
	Mitigation	0.78	0.22	**0.00**	0.10	0.90	0.01

TABLE VI
COT-OUTPUT ALIGNMENT OF FINALIZED COT FORMS ON 80 PAIRED PROMPTS.

Form	Synth Dist. ($\mu \pm \sigma$)	Δ vs. benign	p-value vs. benign
benign CoT	34.75 $\pm$ 4.44	–	–
Hijacked CoT	**22.58** $\pm$ **5.86**	**12.16**	< 0.01
Early-Reflection	32.65 $\pm$ 4.12	2.10	< 0.01
Post-Reflection	27.58 $\pm$ 4.12	7.17	< 0.01

trigger-activated CoT hijacking behavior after implantation.

The mitigation, however, introduces a refusal-side trade-off. In Table VII, MMLU remains close to neighboring settings for Qwen2.5-7B and Llama3.2-3B, and GSM8K stays within a comparable range. At the same time, Table V shows that safe-prompt FR increases relative to the hijacking setting (e.g., DeepSeek-7B: 0.15 $\rightarrow$ 0.34, Qwen2.5-7B: 0.16 $\rightarrow$ 0.30, Llama3.2-3B: 0.14 $\rightarrow$ 0.22), while unsafe-prompt FR remains high. These results indicate a stricter refusal policy: mitigation suppresses hijacked behavior, but it also raises false refusals on some benign prompts, especially those containing safety-related terms. A qualitative case study that contrasts Stage 1, Stage 2, and mitigation under the same trigger is provided in Appendix-F.

C. Ablation Study

1) Impact of Different CoT Hijacking Patterns: To analyze how different forms of CoT affect the attack, we compare five settings that cover Stage-1 mismatch, direct hijacking, reflection-based variants, and mitigation. All settings use the same trigger-based context and differ mainly in the structure of generated CoTs and in whether safety reflection is included. This design lets us examine how CoT format affects both output-level attack behavior and CoT-pattern conformity.

- Output-only hijacking with CoT–output mismatch (S1): the trigger steers the final output to malicious content while the CoT remains benign.
- Direct CoT hijacking (S2): under the trigger, both CoT and final output are malicious.
- Early-Reflection hijacking (S3): the model emits a hijacking flag, then reflects and produces benign CoT while the final output remains malicious.
- Post-Reflection hijacking (S4): the model emits a hijacking flag, then produces partial malicious CoT followed by reflection and benign CoT, while the final output remains malicious.
- Mitigated dual-layer benign CoT (S5): the mitigation setting, reported as a reference point rather than as a hijacking pattern.

First, as shown in Table VII, output-level attack behavior remains strong across multiple settings. In S1 and S2, ASR is high under the trigger, whereas both CHR and ASR remain low without the trigger. For example, on AdvBench at Pass@1, DeepSeek-7B achieves ASR=1.00 in S1 and 0.99 in S2; on StrongREJECT at Pass@1, Qwen2.5-7B achieves ASR=0.94 in S1 and 1.00 in S2. In S3 and S4, ASR often remains substantial even though CHR declines. This indicates that output-level trigger control is not restricted to a single observed CoT format.

Second, S1–S4 show a clear CHR trend as the supervised hijacking pattern becomes more complex. In S1, CHR is 0.00 because the supervised CoTs are intentionally benign. In S2, supervising a fully malicious CoT drives CHR close to 1.00. In S3 and S4, adding structural elements (e.g., hijacking flags and reflection segments) increases compositional complexity and lowers CHR. For example, on StrongREJECT at Pass@1 for Qwen2.5-7B, CHR decreases from 0.74 in S3 to 0.13 in S4.

A plausible explanation is that CHR favors a relatively consistent CoT pattern. When CoTs mix malicious prefixes, reflection steps, and benign segments, generated CoTs distribute across multiple styles rather than concentrating on one dominant pattern. Under this view, lower CHR in S3 and S4 reflects weaker pattern consistency, not necessarily weaker output-level attack behavior.

S5 serves as a mitigation reference. After mitigation, CHR and ASR are suppressed in most settings (e.g., on AdvBench at Pass@1, Qwen2.5-7B has CHR=0.00 and ASR=0.01), while MMLU remains within a narrow range (Qwen2.5-7B: 0.72–0.73; Llama3.2-3B: 0.58–0.59). Overall, the S1–S5 comparison supports two conclusions: trigger-conditioned output control can persist across CoT formats, and observed CoT behavior varies systematically with CoT structure during fine-tuning.

2) Impact of Different Backdoor Data Ratios: We order the settings from 4000 + 80 to 6000 + 80, which corresponds to decreasing effective backdoor-data ratio as the amount of benign training data increases.

Table VIII evaluates how the benign-to-backdoor data composition affects attack behavior when the backdoor set is fixed at 80 samples. At 6000 + 80, all three models show consistently high trigger-conditioned CHR/ASR (0.98–1.00 across benchmarks) and zero off-trigger CHR/ASR in this table. This indicates strong trigger selectivity: the attack remains effective under the trigger while off-trigger activation is effectively suppressed.